UNIT 5 • SYSTEMS OF EQUATIONS

Activity 5.2: Elimination & Word Problems

A

Name: _____ Partner's name: _____ Date: _____

Directions: Solve each system using elimination (or another method). Write answers as ordered pairs for #1–6. Problems 7 and 8 are word problems: define your variables.

Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.

1

$$\begin{cases} x + y = 7 \\ x - y = 3 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

2

$$\begin{cases} 2x + y = 11 \\ 3x - y = 4 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

3

$$\begin{cases} 3x + 2y = 12 \\ 3x - y = 3 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

4

$$\begin{cases} 2x + 3y = 1 \\ 4x - 3y = 11 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

5

$$\begin{cases} x + 2y = 6 \\ 3x + 4y = 16 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

6

$$\begin{cases} 2x + 3y = -4 \\ 3x - 2y = 7 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

7

Adult tickets cost \$8 and student tickets cost \$5. A total of 10 tickets were sold for \$62. How many adult tickets and how many student tickets were sold?

Answer: _____ Match? ☐ Yes ☐ No

8

The sum of two numbers is 20. Their difference is 6. Find the two numbers.

Answer: _____ Match? ☐ Yes ☐ No

Activity 5.2: Elimination & Word Problems

B

Name: _____ Partner's name: _____ Date: _____

Directions: Solve each system using elimination (or another method). Write answers as ordered pairs for #1–6. Problems 7 and 8 are word problems: define your variables.

Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.

1

$$\begin{cases} 2x + y = 12 \\ x - y = 3 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

2

$$\begin{cases} x + 2y = 13 \\ 4x - 2y = 2 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

3

$$\begin{cases} 2x + 3y = 13 \\ 2x - y = 1 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

4

$$\begin{cases} x + 3y = -1 \\ 5x - 3y = 13 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

5

$$\begin{cases} 2x + y = 9 \\ 5x + 3y = 23 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

6

$$\begin{cases} 5x + 2y = 1 \\ 2x - 3y = 8 \end{cases}$$

Answer: _____ Match? ☐ Yes ☐ No

7

A café sells bagels for \$4 and juices for \$2. A customer buys 10 items for \$28. How many bagels and how many juices did the customer buy?

Answer: _____ Match? ☐ Yes ☐ No

8

Two numbers have an average of 10. The larger number is 6 more than the smaller number. Find the two numbers.

Answer: _____ Match? ☐ Yes ☐ No